

Integration and Configuration Documentation for OpenEvolve Symbolic Regression

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October 24, 2025

Overview

Current Progress and Minimal Integration Changes

I integrated the symbolic regression workflow within the OpenEvolve framework. I created a new symbolic target in the Makefile to enable iterations of regression benchmarks using Docker. This target links the evaluator and initial program scripts with a configurable iteration count, ensuring consistent experimentation across different directories. I also added throttling controls after encountering a rate limit per minute, including a short delay between requests to manage the load. In addition, I wrote a small Bash script that iterates through the problems within symbolic regression.

Moving forward, I'll be able to update my changes as I work through each problem, making them easier to track. I accidentally uploaded my work after editing everything, so it's hard to tell exactly what I changed—but this setup will make future updates much clearer.

1. Makefile Updates

A new target, `symbolic`, was added to the Makefile to automate symbolic regression runs. This target ensures consistency and reproducibility when launching Docker-based benchmarks.

```
.PHONY: symbolic
symbolic:
    docker run --rm -v $(PROJECT_DIR):/app --network="host" $(DOCKER_IMAGE) \
        examples/symbolic_regression/bench/initial_program.py \
        examples/symbolic_regression/bench/evaluator.py \
        --config examples/symbolic_regression/bench/config.yaml \
        --iterations 100
```

Purpose: This addition standardizes the symbolic regression benchmark process, reducing manual setup and ensuring reproducible experiments. The iteration parameter remains configurable via the Makefile.

2. Automated Bash Script for Multi-Domain Reporting

Created a Bash script was to improve the legibility of output logs and for consistent comparisons for other symbolic regression experiments.

```
#!/bin/bash
set -e

BASE_DIR="examples/symbolic_regression/problems"
DOMAINS=("bio_pop_growth" "chem_react" "matsci" "phys_osc")

echo "{"
echo '  "OpenEvolve_Run": {'
echo '    "start_time": "'$(date '+%Y-%m-%d %H:%M:%S')'",'
echo '    "domains": ['

for domain in "${DOMAINS[@]"; do
  DOMAIN_PATH="$BASE_DIR/$domain"
  if [ -d "$DOMAIN_PATH" ]; then
    echo "      {"
    echo '        "domain": "'$domain'",'
    echo '        "problems": ['

    for problem_dir in "$DOMAIN_PATH"/*; do
      [ -d "$problem_dir" ] || continue
      PROBLEM_NAME=$(basename "$problem_dir")
      LOG_PATH="$problem_dir/openevolve_output/logs/openevolve_$(date '+%Y%m%d').log"

      echo '          {'
      echo '            "problem": "'$PROBLEM_NAME'",'
      echo '            "metrics": {'
      echo '              "combined_score": null,'
      echo '              "negative_mse": null,'
      echo '              "complexity": null,'
      echo '              "diversity": null'
      echo '            },'
      echo '            "log_path": "'$LOG_PATH'"'
      echo '          },'
    done

    echo '        ],'
  fi
done

echo '    ],'
echo '  },'
echo '}"
```

3. Configuration File Modifications

```
# Throttling controls
rate_limit_per_minute: 1    # Only 1 request per minute
sleep_between_requests: 5    # Wait 5 seconds between API calls
```
